

Solution Summary

Sentinel, a unified video management and cross camera vehicle intelligence system for Gujarat Police

PROBLEM	26 department camera estates that cannot see each other
APPROACH	Edge analysis, metadata only transport, geography as filter
DELIVERED	Working system, 386 automated tests passing
DOCUMENT	3 of 3, Solution Summary

The problem, and the approach

The problem, stated plainly

CCTV in Gujarat belongs to 26 separate police departments and municipal bodies, and none of them can see the cameras of another. When a stolen vehicle drives from Ahmedabad to Surat it crosses an institutional boundary and the trail stops there. Not because the cameras are missing, but because the systems behind them were never joined. Sentinel joins these estates into one investigation screen without replacing a single installed camera.

Three engineering decisions shape the whole design

1

Video does not travel. Only information travels.

Sending 80,000 camera streams to one location needs about 320 Gbps of sustained backhaul. That is not a budget problem, it is a physical one. So analysis happens at the edge and only a 400 byte record is forwarded: plate, vehicle type, colour, time, camera, direction. Statewide that is about 96 Mbps, roughly 3,300 times smaller. Video stays in the district that owns it.

2

Geography does the filtering, not the neural network.

A directional road network graph decides which cameras a vehicle could physically have reached in the elapsed time. Candidates fall from about 20,000 to 473, a reduction of 97.6 percent, and matching falls from 2,240 ms to 53 ms. The point is accuracy rather than speed: a 99 percent accurate matcher produces roughly 200 wrong links an hour against 20,000 candidates and about four against 473.

3

The system states when it is unsure.

Only about 26 percent of cameras in a real estate can physically resolve a number plate. The rest are wide angle, which is physics and not a software gap. So appearance is also matched, but a rule in the code prevents appearance evidence alone from ever returning a confirmed identification. Confirmed requires a plate read. Every link carries the evidence behind it.

CONSEQUENCE

The architecture those three decisions produce

Three tiers. Edge nodes handle about 40 cameras each and video terminates there permanently, roughly 500 of them for the full state. Regional nodes correlate across district boundaries, because a vehicle crossing one is exactly the case a single department cannot see. A central site at Gandhinagar holds the map, the search, the user accounts and the audit trail. Every arrow points upward: there is no path by which the centre can reach into a district and pull video on its own initiative.

Protocol agnostic by requirement, not by ambition

Analog cameras on a DVR, modern IP cameras and municipal HLS feeds are all brought online by editing one configuration file, with no code change and no rebuild. Where a government firewall blocks the RTSP port, which is common, the reader rotates to HLS automatically rather than reporting the camera as dead and filling an operator screen with false alarms.

Key features

Every item below is implemented in the delivered system and covered by the automated test suite.

Tracking and intelligence

- Cross camera vehicle tracking across 26 independent department estates
- Road network filtering that removes 97.6 percent of candidate comparisons
- Number plate reading with Indian plate grammar and ambiguous character rules
- Appearance matching for the cameras that cannot resolve a plate
- A hard rule that appearance alone can never return a confirmed match
- Match reasoning shown to the officer, not only a confidence number

Ingestion and resilience

- Edge processing, so video never leaves the district that owns it
- Automatic switch from RTSP to HLS when a firewall blocks the port
- Timing taken from the video container rather than from packet arrival
- Stream break detection that resets the tracker and clears stale data
- Analog, IP and municipal cameras onboarded through one configuration file
- Database replication with WAL archiving and point in time recovery

Integration and interfaces

- 42 REST endpoints across auth, cameras, vehicles, alerts, intelligence, system
- Vehicle search by full plate, partial plate, colour, type and time window
- Alerts for watchlist hits, restricted zone entry and cameras going offline
- Adapters prepared for VAHAN, SARTHI, eGujCop, AFIS and NAFIS
- Prometheus metrics and Grafana dashboards for operations

Security, privacy and operations

- Department isolation enforced in the database, not in application code
- A stated reason required for every person linked lookup, kept seven years
- Audit records that cannot be quietly edited; reading the audit is audited
- Camera credentials structurally excluded from the database
- Six roles from system administrator down to read only viewer
- Account lockout with an audited operator recovery path
- Live command centre map with real time updates over WebSocket
- One command deployment with Docker Compose, Kubernetes for production

Expected impact

Measured on the delivered system

What	Result	Verified by
Statewide network required	320 Gbps to 96 Mbps	Bandwidth plan
Comparisons per sighting	20,000 to 473	Benchmark suite
Cross camera match latency	2,240 ms to 53 ms	Benchmark suite
Department scoped query	38 ms to 0.72 ms	Isolation regression
Records lost on primary failure	Zero	Failover drill
Time to accept writes again	0.36 seconds	Failover drill
Security findings closed	18 of 18	Security review
Cross department data leaks	Zero	Isolation suite
Automated tests passing	386 of 386	Live PostgreSQL run

What this changes in practice

Investigation time falls from days to minutes

Following a vehicle across two districts currently means telephoning another department, requesting footage, waiting, and reviewing it by hand. That is one to three days realistically, and it is often abandoned. Sentinel returns the cross district history in seconds, timestamped and attributable, so it still holds up when the case reaches a court.

Existing infrastructure is preserved

No camera is replaced. No department loses custody of its own video. Joining the system costs a configuration file rather than a procurement cycle.

Wrong identification is constrained by design

Because appearance evidence cannot be promoted to a confirmed match, the system is structurally incapable of presenting a colour and shape guess as a positive identification. The published accuracy range is deliberately honest: 92.5 percent on a dedicated plate reading lane in daylight, 37.9 percent on a wide angle camera at night. Publishing the poor figure is what tells an officer when not to act.

Cost position, stated as modelling and not as a quotation

Our modelling puts the edge based approach at about 82.7 crore rupees against about 138.7 crore rupees for conventional video centralisation in year one, a saving near 40 percent, almost all of it from not building 320 Gbps of backhaul. These are estimates against list prices. No procurement quotation has been obtained.

What is not done, stated before it is asked

A reviewer who discovers an unstated gap discounts every other claim in a submission. The list below is therefore given up front, so that the proven work can be weighed on its own merits.

Not done	Why, and what it depends on
Connection to the live Gujarat camera grid	Test cameras were provided, not the production estate
VAHAN, SARTHI, eGujCop, AFIS, NAFIS live endpo	Adapters are written and tested against mocks; each needs its own
Accuracy measured on real Gujarat footage	Present figures come from the simulation model, not from field video
Validation above 1,000 cameras	Beyond that point the figures are calculated rather than measured
Continuous integration pipeline	The suite runs locally with one command; CI is a known next step

IMPORTANT, AND EASY TO MISREAD

On the AI models specifically

Detection runs on ONNX Runtime and accepts YOLO or RT DETR exports, choosing GPU or CPU at runtime. Re identification uses 512 dimension embeddings sized to OSNet AIN so trained weights drop in with no schema change. For Phase 1 the inference backends run in a deterministic simulation mode that reproduces the real failure modes, including character confusion between O and zero, degradation with plate size and lighting, and outright read failures. This is a deliberate choice: it lets the entire system be evaluated end to end on any machine with no GPU and no downloaded weights. Switching to trained models is one configuration setting.

POSITION

What that leaves as the actual contribution

Anyone can download a detection model, so no submission wins on that. The hard parts of this problem are the road network gate that makes matching trustworthy, the refusal to over claim confidence, the department isolation that lets 26 organisations share one system safely, and a recovery path proven by killing a database rather than by describing a procedure. Those are built, tested and reproducible.

Everything in this submission can be reproduced on request

The failover result comes from a script that builds a primary and a standby, writes 5,000 rows, kills the primary without a clean shutdown and counts what survived. The scale figures come from a load test script. The full suite runs against a live PostgreSQL with one command. No measured figure in these documents depends on a number we cannot regenerate in front of a reviewer.

ASK

What we are asking for in the next phase

Read access to a small block of production cameras in one district, so accuracy can be measured on Gujarat footage rather than modelled. An introduction to the authorities behind VAHAN, SARTHI and eGujCop, since the adapters are written and only the approvals are outstanding. And a nominated officer to sit with the command centre for a day, because the interface should be judged by the person who will actually use it at three in the morning, not by the people who built it.